POWER BI ASSIGNMENT-2

Infanta Kennedy

ENTRI

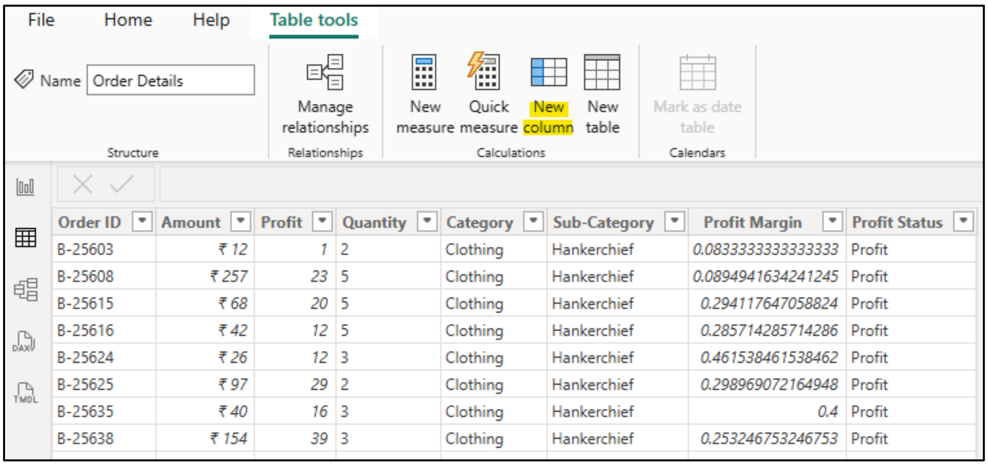
DAX & Data Visualization

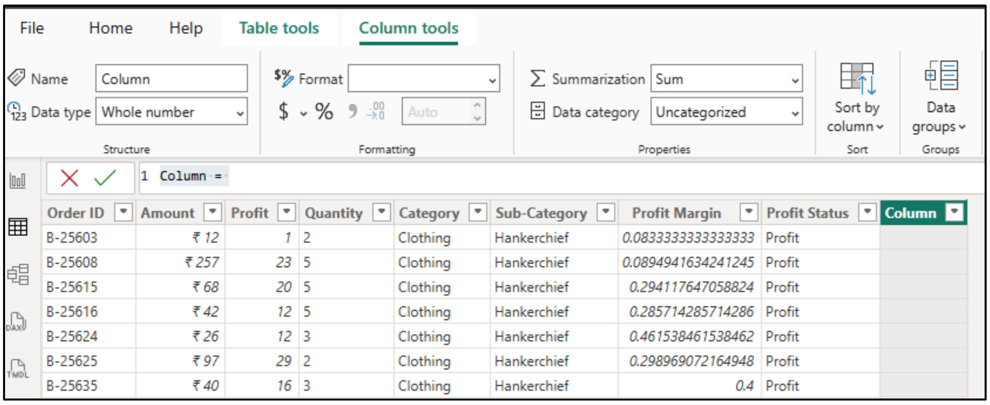
Calculated Columns

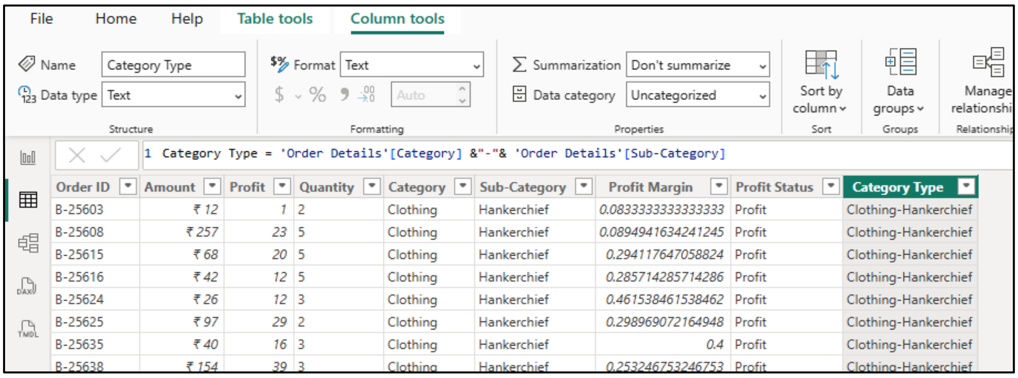
● Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Steps:

1. Open Power BI Desktop
2. Go to the Table View
3. Select the Order Details table
4. Click New Column
5. Type the DAX formula - Category Type = 'Order Details'[Category]&"-"&'Order Details'[Sub-Category]
6. Press Enter





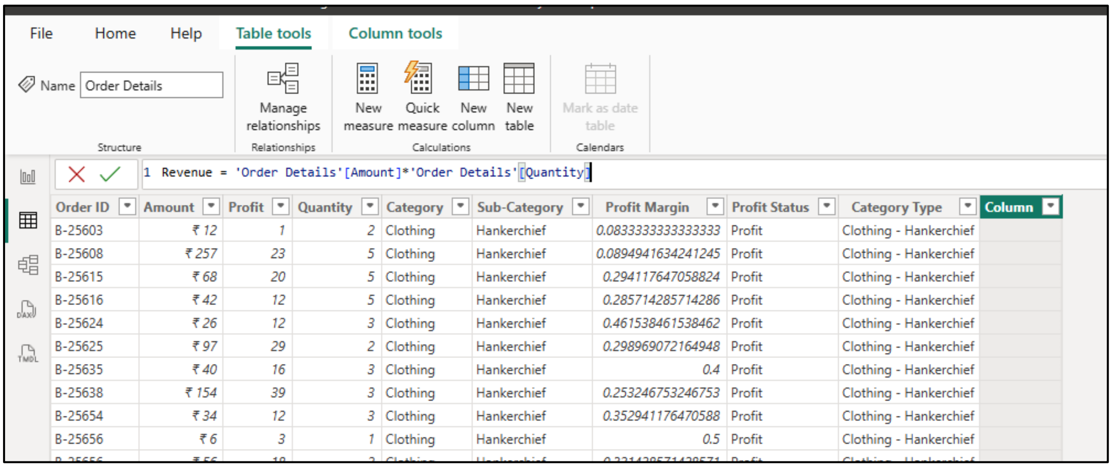


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing-Hankerchief
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing-Hankerchief
B-25615	₹ 68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing-Hankerchief
B-25616	₹ 42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing-Hankerchief
B-25624	₹ 26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing-Hankerchief
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing-Hankerchief
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief
B-25638	₹ 154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing-Hankerchief

● Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

Steps:

1. Go to Table View
2. Select the Order Details table
3. Click New Column
4. Type the DAX formula - Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]
5. Press Enter



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Column
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	
B-25615	₹ 68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	
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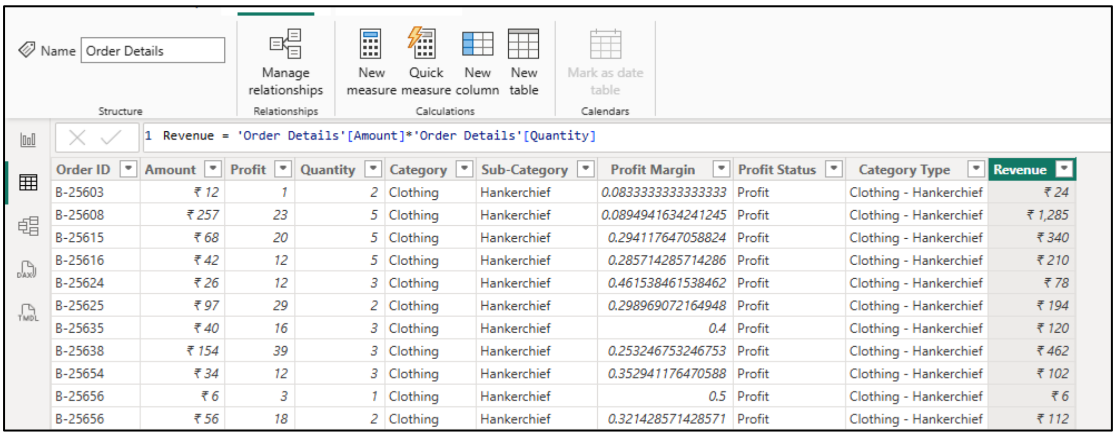
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	
B-25638	₹ 154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	
B-25654	₹ 34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	
B-25656	₹ 6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	

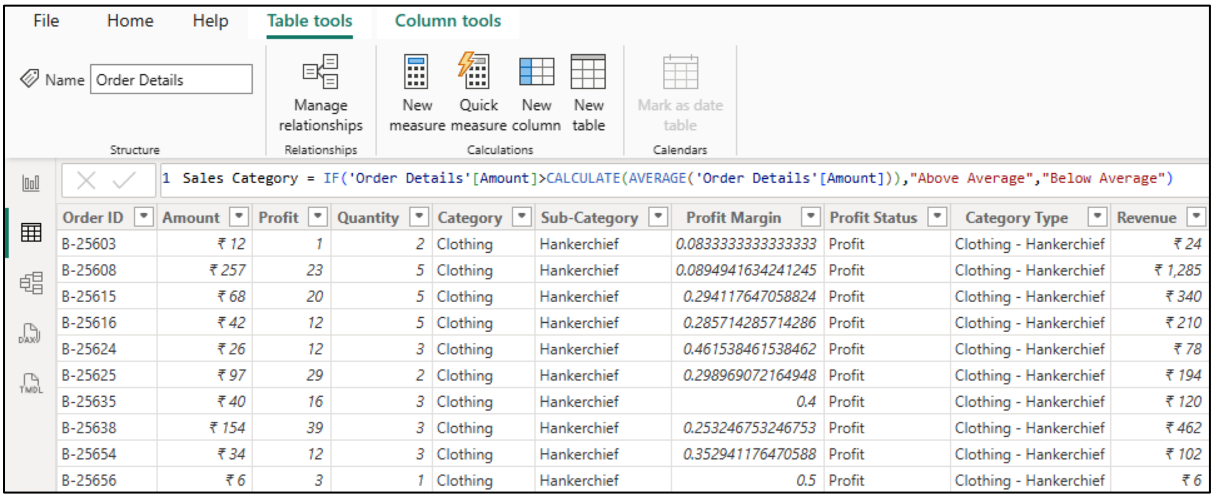


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	₹ 24
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285
B-25615	₹ 68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	₹ 340
B-25616	₹ 42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief	₹ 210
B-25624	₹ 26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief	₹ 78
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	₹ 194
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120
B-25638	₹ 154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	₹ 462
B-25654	₹ 34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	₹ 102
B-25656	₹ 6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6
B-25656	₹ 56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief	₹ 112

● Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

Steps:

1. Select Order Details table in table view
2. Click → New Column
3. Type the DAX formula - Sales Category = IF('Order Details'[Amount]>CALCULATE(AVERAGE('Order Details'[Amount])),"Above Average","Below Average")
4. Press Enter



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue
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B-25608	₹ 257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285
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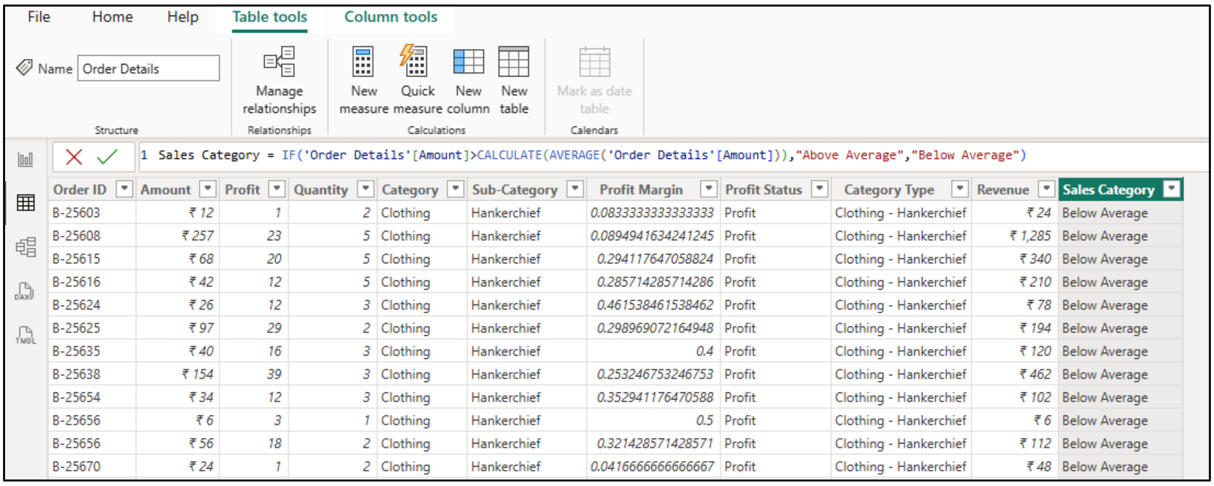
B-25656	₹ 6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
B-25603	₹ 12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing - Hankerchief	₹ 24	Below Average
B-25608	₹ 257	23	5	Clothing	Hankerchief	0.0894941634241245	Profit	Clothing - Hankerchief	₹ 1,285	Below Average
B-25615	₹ 68	20	5	Clothing	Hankerchief	0.294117647058824	Profit	Clothing - Hankerchief	₹ 340	Below Average
B-25616	₹ 42	12	5	Clothing	Hankerchief	0.285714285714286	Profit	Clothing - Hankerchief	₹ 210	Below Average
B-25624	₹ 26	12	3	Clothing	Hankerchief	0.461538461538462	Profit	Clothing - Hankerchief	₹ 78	Below Average
B-25625	₹ 97	29	2	Clothing	Hankerchief	0.298969072164948	Profit	Clothing - Hankerchief	₹ 194	Below Average
B-25635	₹ 40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing - Hankerchief	₹ 120	Below Average
B-25638	₹ 154	39	3	Clothing	Hankerchief	0.253246753246753	Profit	Clothing - Hankerchief	₹ 462	Below Average
B-25654	₹ 34	12	3	Clothing	Hankerchief	0.352941176470588	Profit	Clothing - Hankerchief	₹ 102	Below Average
B-25656	₹ 6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing - Hankerchief	₹ 6	Below Average
B-25656	₹ 56	18	2	Clothing	Hankerchief	0.321428571428571	Profit	Clothing - Hankerchief	₹ 112	Below Average
B-25670	₹ 24	1	2	Clothing	Hankerchief	0.0416666666666667	Profit	Clothing - Hankerchief	₹ 48	Below Average

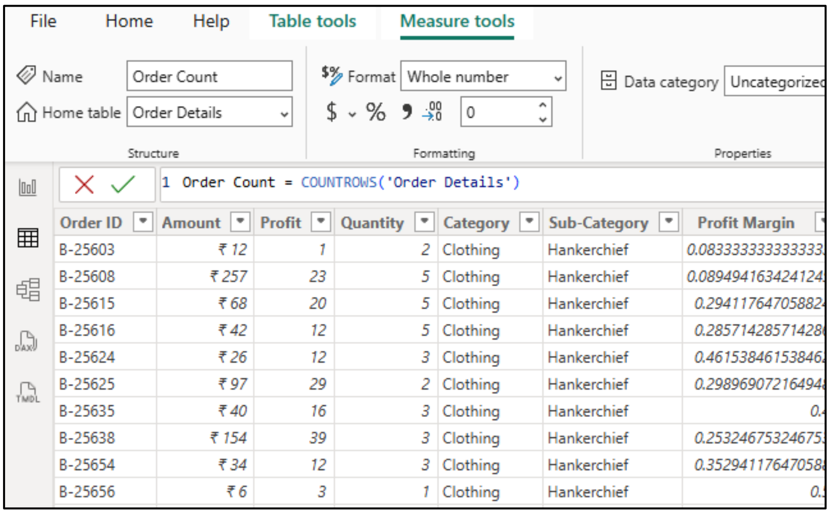
Calculated Measures:

● Calculate Order Count: Define a measure to count the total number of orders in the Order Details table.

Steps in Microsoft Power BI Desktop

1. Select the Order Details table
2. Go to Table Tools -> New Measure
3. Type the formula - Order Count = COUNTROWS('Order Details')
4. Press Enter

- COUNTROWS() counts the total number of rows/orders in the Order Details table.



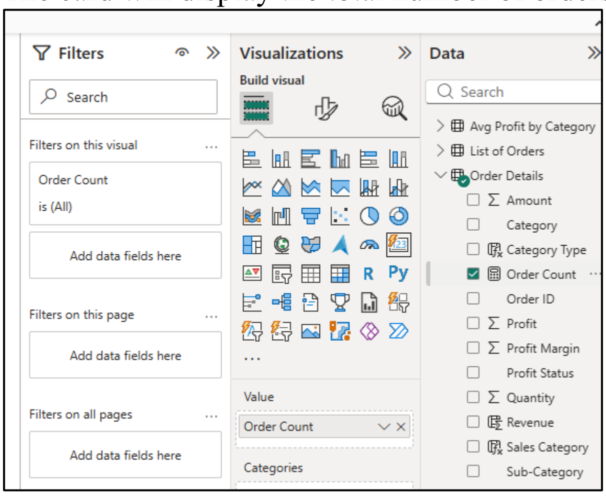
To display the Order Count measure in Microsoft Power BI Desktop:

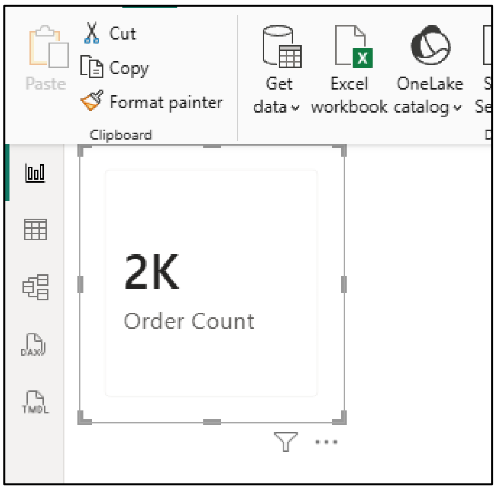
Steps

1. Go to the Report View
2. In the Visualizations pane, select a Card visual
3. Drag the Order Count measure from the Fields pane into the card

Result

The card will display the total number of orders in the Order Details table.





● Calculate Average Profit in Delhi: Create a measure to calculate the average profit for orders placed in Delhi.

As City Delhi is not available in given dataset, I have used Chennai city for the analysis

Steps:

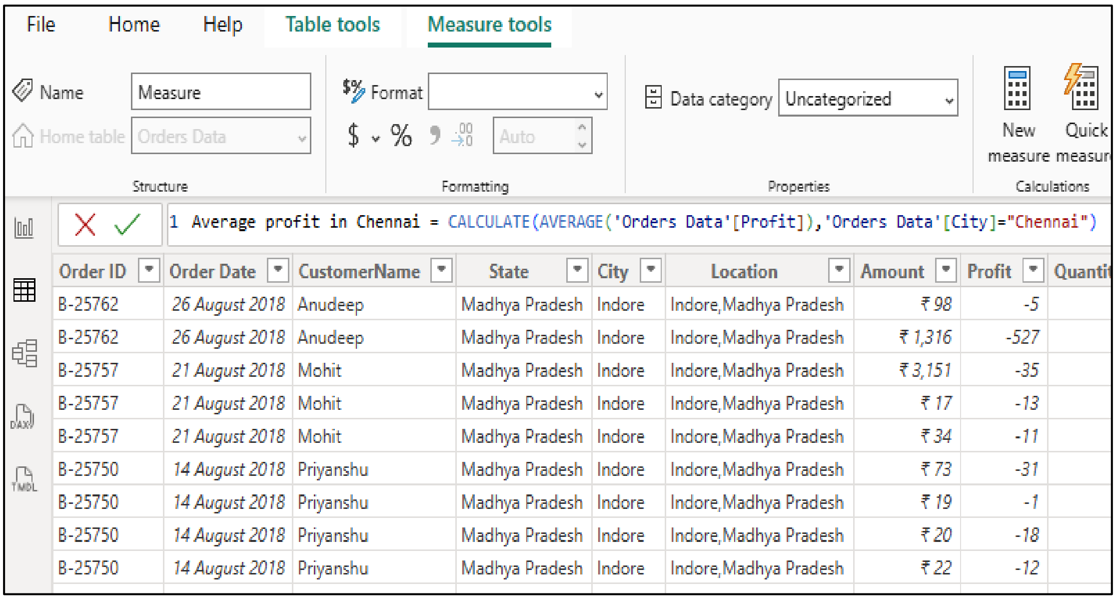
1. Select the Orders Data table
2. Click Table Tools → New Measure
3. Type the formula - Average profit in Chennai = CALCULATE(AVERAGE('Orders Data'[Profit]),'Orders Data'[City]="Chennai")
4. Press Enter

- AVERAGE() calculates the average profit
- CALCULATE() filters the data only for orders where City = Chennai

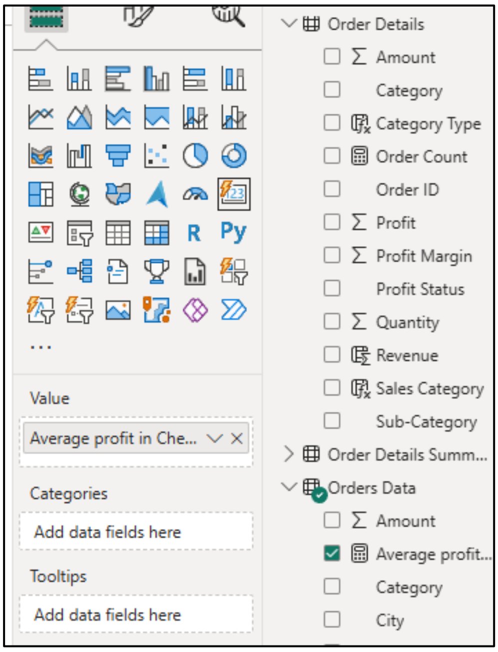
To Display the Result

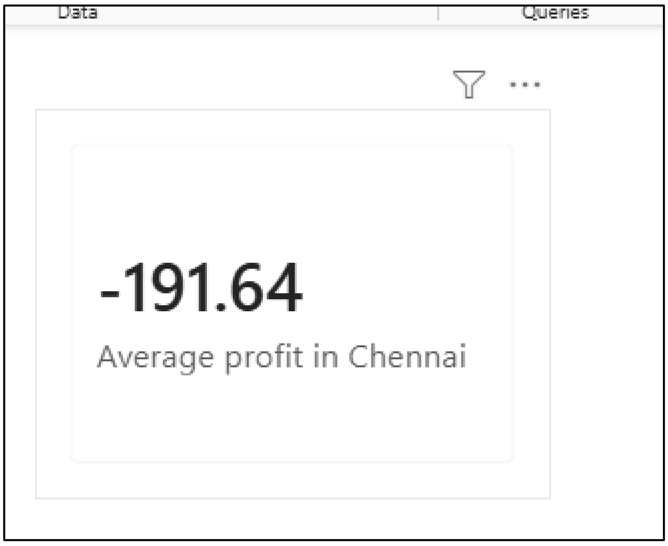
- Add a Card visual
- Drag Average Profit Chennai measure into the card

This will show the average profit for Chennai orders only.



Order ID	Order Date	CustomerName	State	City	Location	Amount	Profit	Quantit
B-25762	26 August 2018	Anudeep	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 98	-5	
B-25762	26 August 2018	Anudeep	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 1,316	-527	
B-25757	21 August 2018	Mohit	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 3,151	-35	
B-25757	21 August 2018	Mohit	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 17	-13	
B-25757	21 August 2018	Mohit	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 34	-11	
B-25750	14 August 2018	Priyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 73	-31	
B-25750	14 August 2018	Priyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 19	-1	
B-25750	14 August 2018	Priyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 20	-18	
B-25750	14 August 2018	Priyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh	₹ 22	-12	





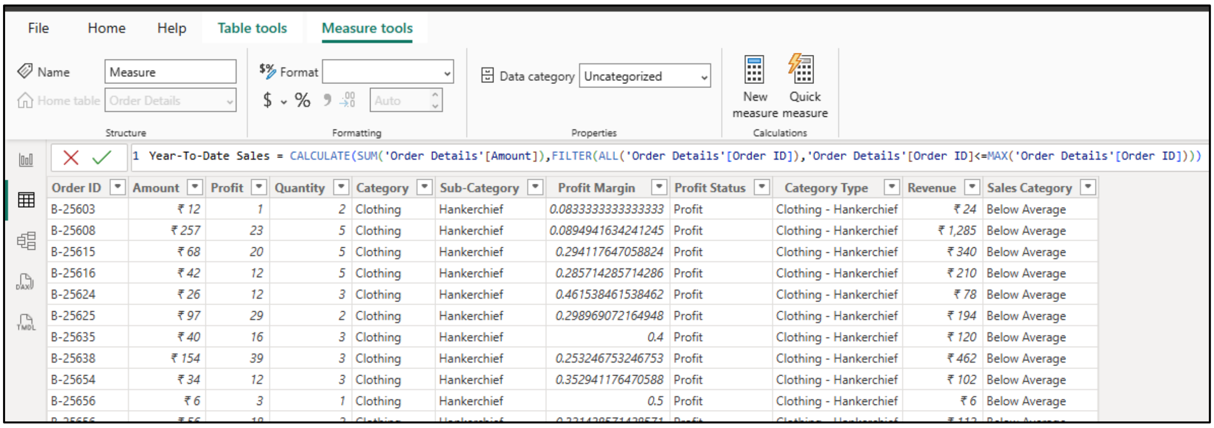
● Calculate Year-to-Date (YTD) Sales: Define a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

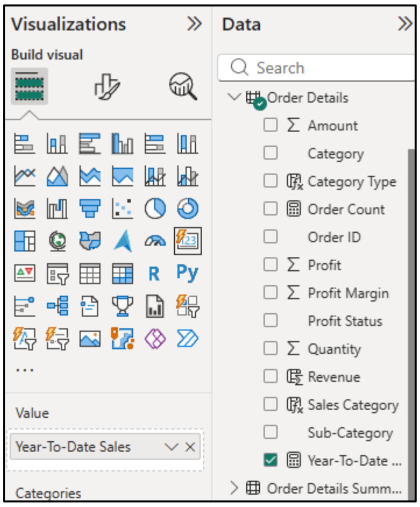
Steps:

1. Go to Table View
2. Click New Measure
3. Type the DAX formula - Year-To-Date Sales = CALCULATE(SUM('Order Details'[Amount]),FILTER(ALL('Order Details'[Order ID]),'Order Details'[Order ID]<=MAX('Order Details'[Order ID])))
4. Press Enter

What this measure does

- SUM(Amount) → calculates total sales
- MAX(Order Date) → gets the current order date in context
- FILTER() → includes all dates from the earliest date up to the current date
- CALCULATE() → applies the filter context



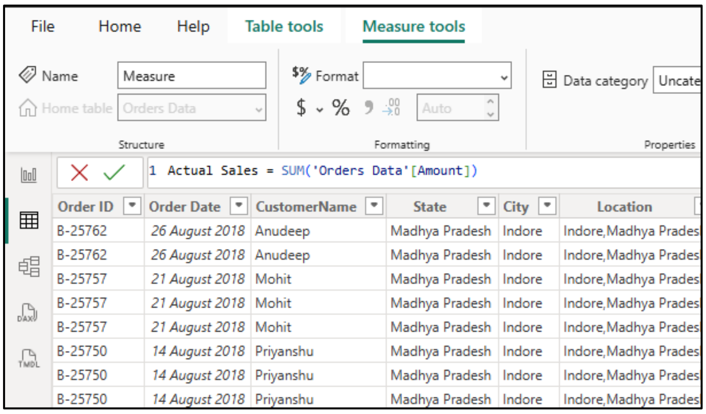


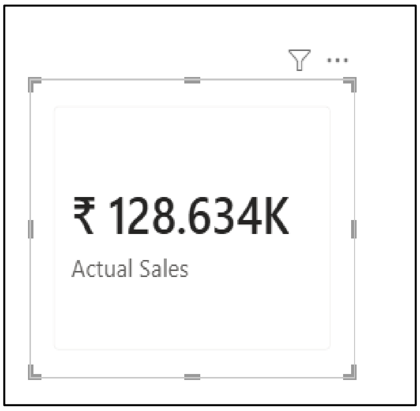
Data Visualization:

- Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.

Step 1: **Creating Actual Sales Measure**

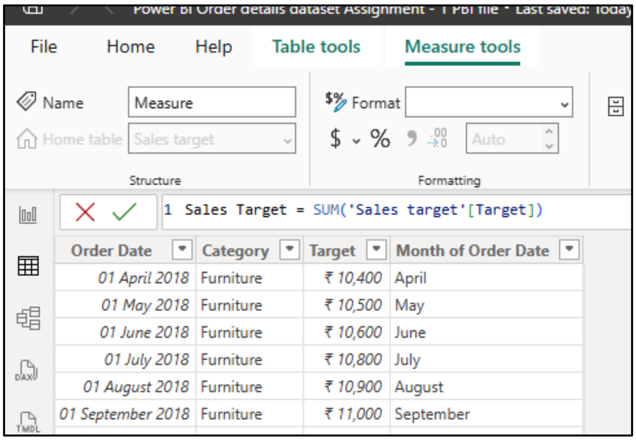
Formula - Actual Sales = SUM('Orders Data'[Amount])

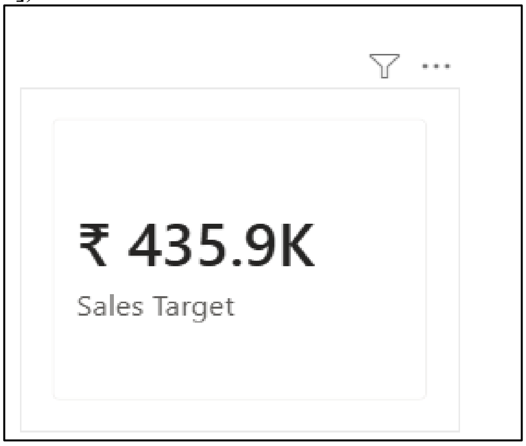




Step 2: Creating Sales Target Measure

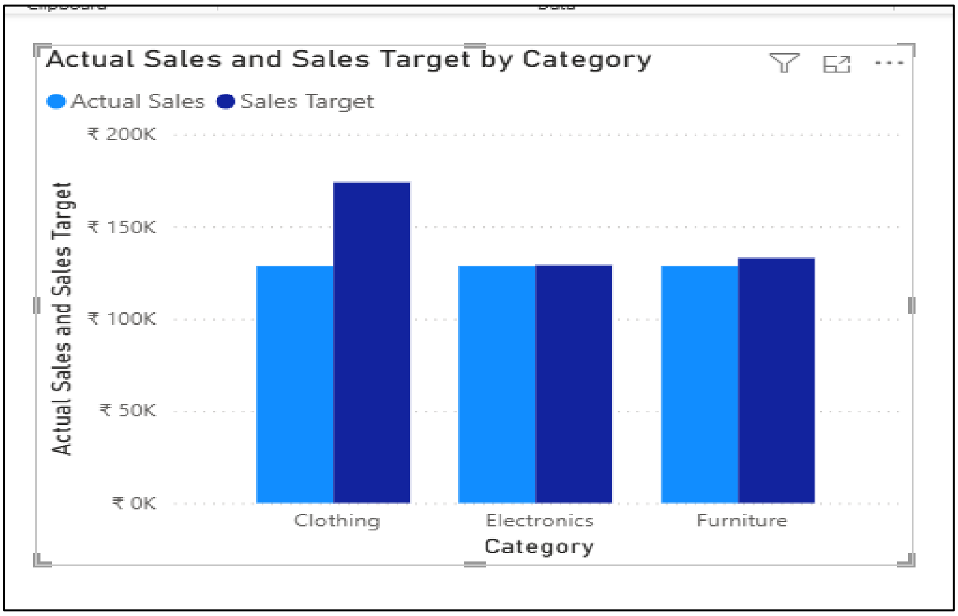
Formula - Sales Target = SUM('Sales target'[Target])





Step 3: Create Clustered Column Chart

1. Go to Report View
2. Select Clustered Column Chart
3. Drag:
    - Category → X-axis
    - Actual Sales → Y-axis
    - Sales Target → Y-axis

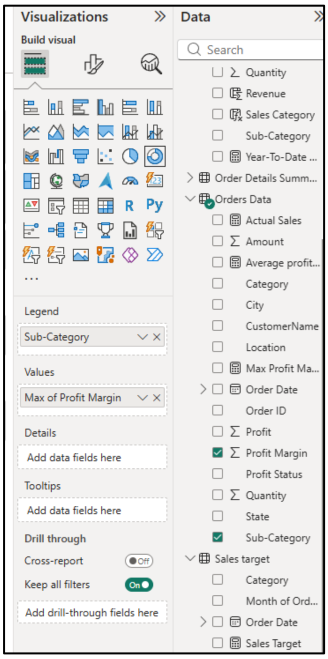


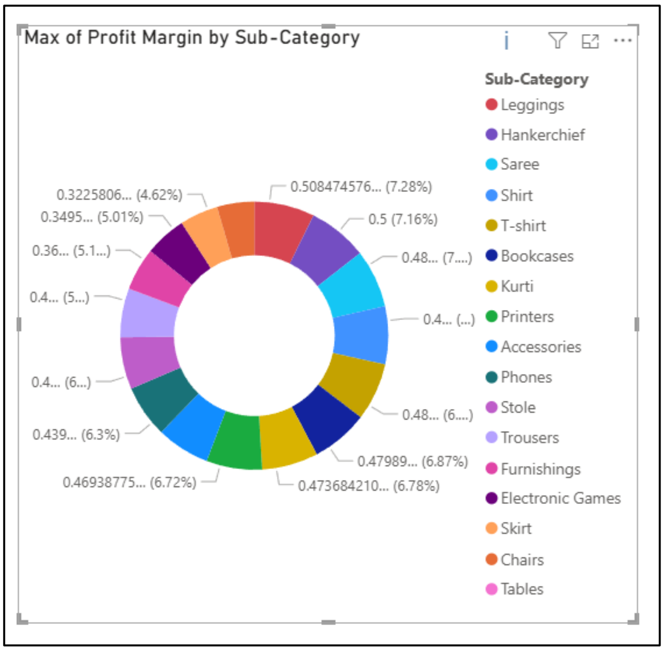
This helps visualize whether each category achieved its sales target.

- Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.

  1. Select Donut Chart
  2. Drag:
     - Sub-Category → Legend
     - Profit Margin column → Values
  3. In the Values dropdown:

     ➔ Change aggregation to Max





This directly shows the maximum profit margin by sub-category.

- Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.

Step 1: Creating a Line Chart

1. Go to Report View
2. Select Line Chart from the Visualizations pane

Step 2: Add Fields

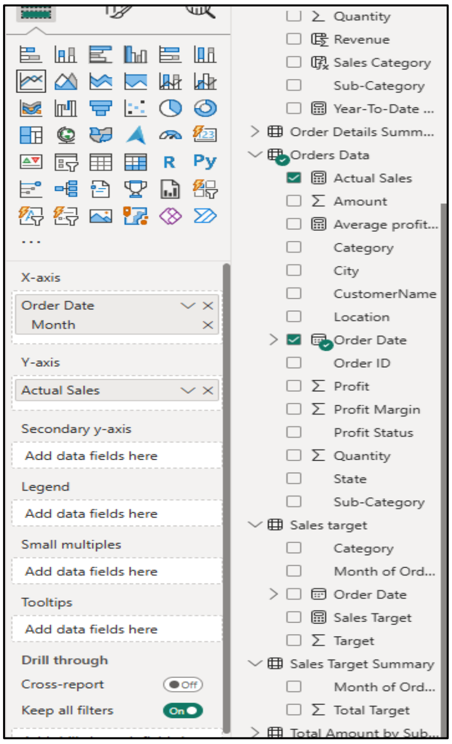
Drag these fields:

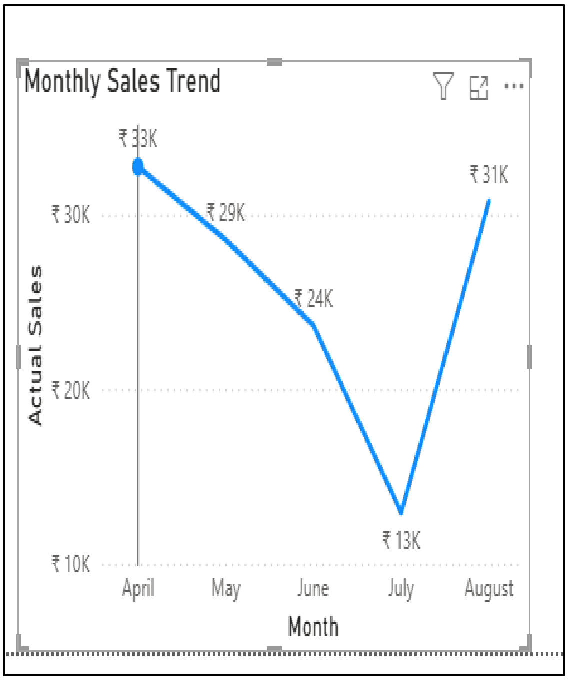
- **Order Date** → X-axis
- **Monthly Sales** → Y-axis

Step 3: Show Month-wise Trend

Then:

1. Click the dropdown beside Order Date
2. Select - Month





- Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

Step 1: Create these two measures

Total Profit = SUM('Orders Data'[Profit])
Total Quantity = SUM('Orders Data'[Quantity])

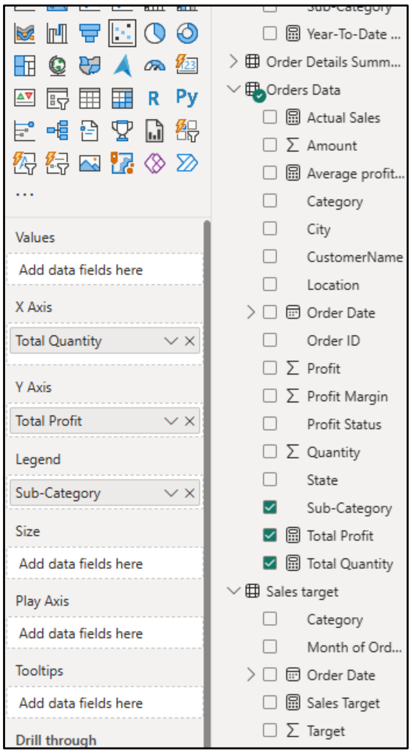
Step 2: Create Scatter Chart

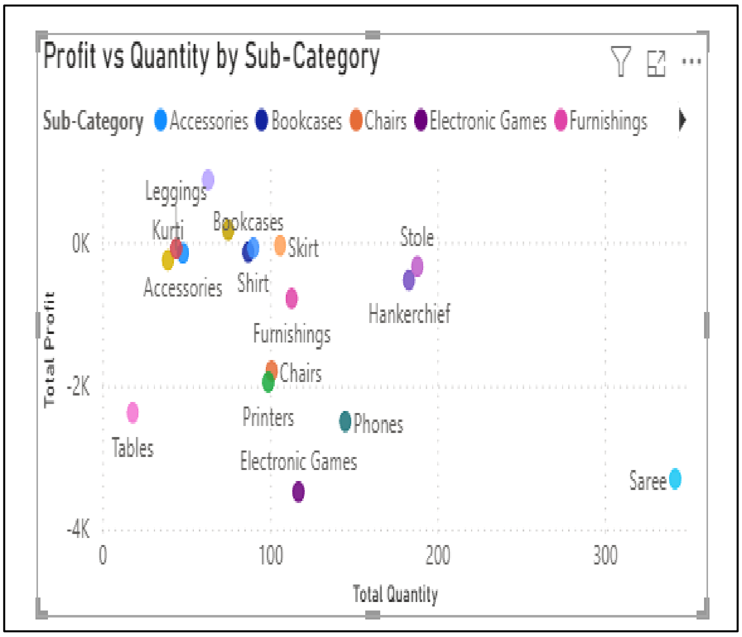
1. Go to Report View
2. Select Scatter Chart from the Visualizations pane

Step 3: Add Fields

Drag these fields into the scatter chart:

- **Total Quantity** → X-axis
- **Total Profit** → Y-axis
- **Sub-Category** → Legend





Helps identify:

- High-sales & high-profit products
- High-sales but low-profit products
- Low-performing sub-categories

- Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

Creating Cards

Card 1 → Total Sales Amount

1. Select **Card** visual
2. Drag: Total Sales → Fields

Card 2 → Sales Target

1. Add another **Card** visual
2. Drag: Sales Target → Fields

This allows quick comparison between:

- Actual Sales
- Target Sales

- Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

Creating Matrix Visual

In Microsoft Power BI Desktop:

1. Go to **Report View**
2. Select the **Matrix** visual from the Visualizations pane

Steps: Add Fields

Rows - Category

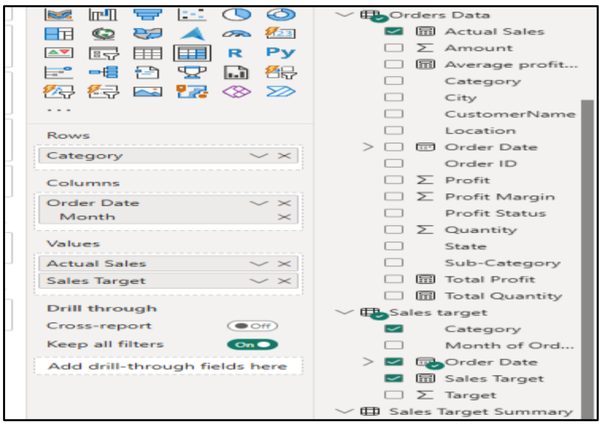
Columns - Order Date

Then:

1. Click dropdown beside Order Date
2. Select: Month

Values - Drag:

- Actual Sales
- Sales Target



Month / Category	January Actual Sales	January Sales Target	February Actual Sales	February Sales Target	March Actual Sales	March Sales Target	April Actual Sales	Sales T
Clothing	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹ 1
Electronics	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹ 16,000	₹ 1,28,634	₹
Furniture	₹ 1,28,634	₹ 11,500	₹ 1,28,634	₹ 11,600	₹ 1,28,634	₹ 11,800	₹ 1,28,634	₹ 1
Total	**₹ 1,28,634**	**₹ 43,500**	**₹ 1,28,634**	**₹ 43,600**	**₹ 1,28,634**	**₹ 43,800**	**₹ 1,28,634**	**₹ 3**

- Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.

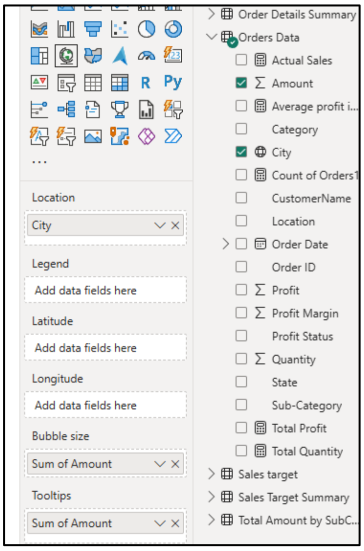
Step 1: Creating Map Visualization

1. Go to **Report View**
2. Select the **Map** visual from the Visualizations pane

Step 2: Add Fields

Drag these fields:

Field	**Place In**
City	Location
Total Sales	Bubble Size / Size
Total Sales	Tooltips



Result

The map will:

- Show cities as bubbles
- Larger bubbles = higher sales
- Smaller bubbles = lower sales

This helps identify:

- High-performing regions
- Regional sales patterns
- Sales concentration by city

- Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a tree map.

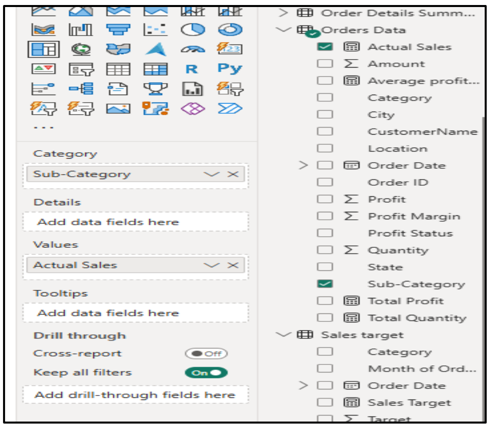
Step 1: Creating Tree map Visual

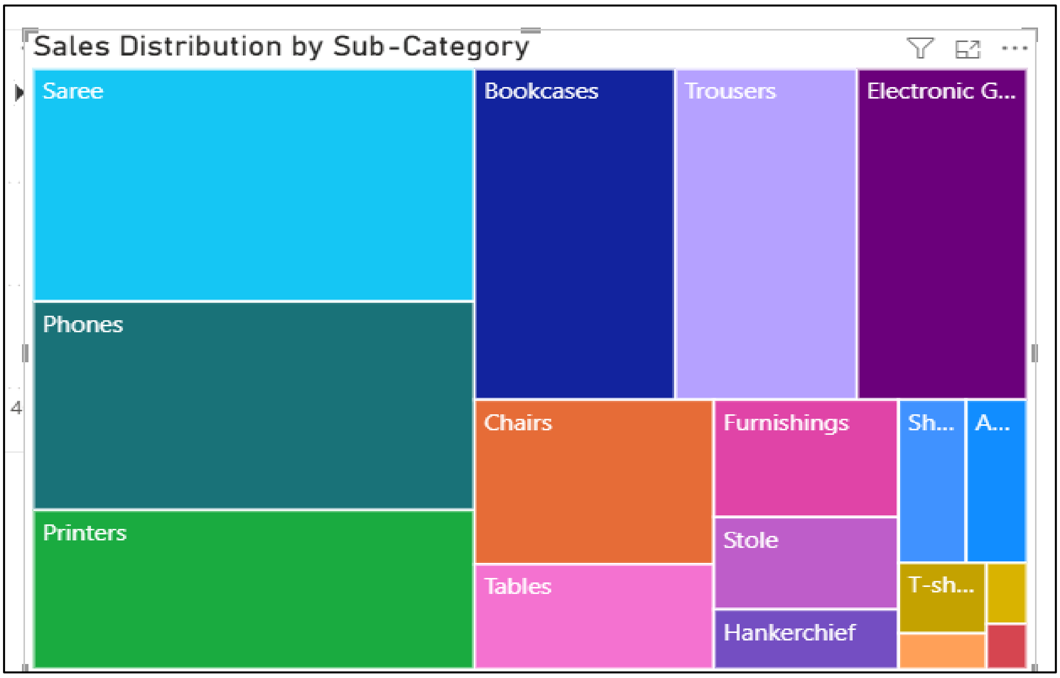
1. Go to Report View
2. Select the Treemap visual from the Visualizations pane

Step 2: Add Fields

Drag these fields:

Field	Place In
Sub-Category	Category
Actual Sales	Values





Result

The Tree map will:

- Display each sub-category as a rectangle
- Rectangle size represents total sales
- Larger rectangles = higher sales contribution

This helps analyse:

- Best-selling sub-categories
- Sales distribution
- Contribution of each product sub-category

- Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.

Step 1: Creating measure for order count

Formula - Count of Orders1 = COUNT('Orders Data'[Order ID])

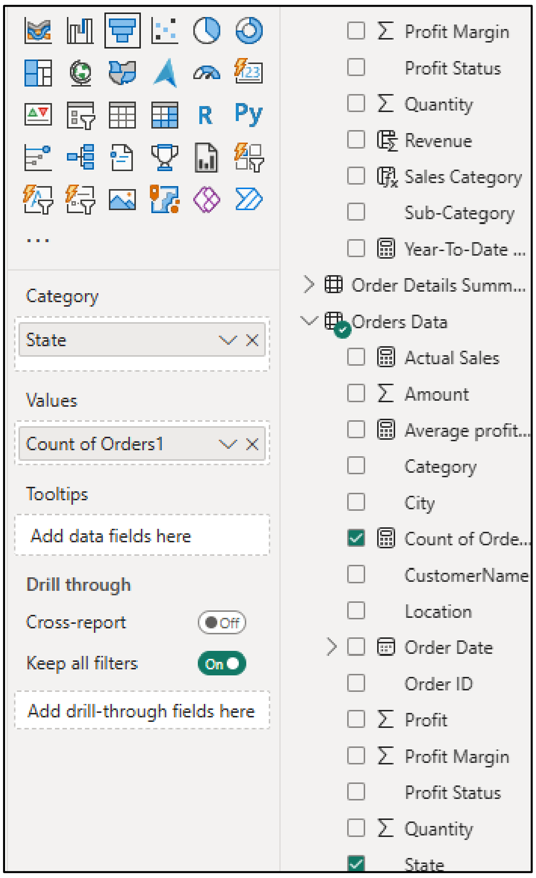
Step 2: Create Funnel Chart

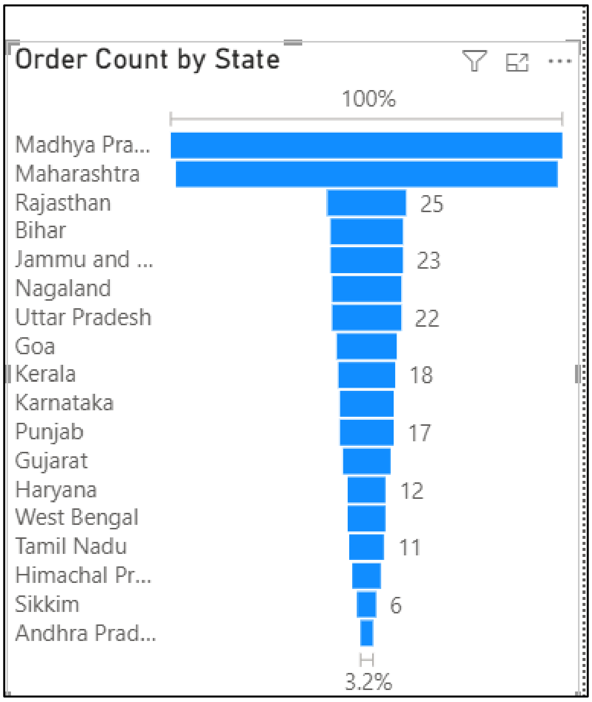
1. Go to Report View
2. Select the Funnel Chart visual from the Visualizations pane

Step 3: Add Fields

Drag these fields:

Field	Place In
State	Category
Order Count	Values





Result

The funnel chart will:

- Display states based on order counts
- Larger sections = more orders
- Smaller sections = fewer orders

This helps identify:

- Top-order states
- Low-performing regions
- Order distribution across states